

Final Project

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Abstract:

This project analyzes daily Uber trip counts in New York City over the 2025 calendar year using data sourced from the NYC Taxi and Limousine Commission (TLC). We apply two time series modeling approaches: a Seasonal ARIMA (SARIMA) model to capture the strong weekly seasonality in ridership demand, and an ARMAX model that extends SARIMA with external regressors including average fare, trip miles, and trip duration. Our results show that both models capture the weekly seasonal pattern well, though residual diagnostics indicate remaining autocorrelation structure. The ARMAX model achieves a lower MAPE than the baseline SARIMA, suggesting that trip-level economic variables provide meaningful predictive signal for daily ridership volume.

Introduction:

Ride-hailing platforms such as Uber have become a central component of urban transportation infrastructure, particularly in high-density cities like New York. Understanding the temporal patterns of ridership demand is valuable for platform operations, driver supply planning, and urban mobility research. In this project, we study daily Uber trip counts in New York City across the full 2025 calendar year, with the goal of modeling and forecasting demand using classical time series methods.

The data exhibits a clear weekly seasonal pattern, with ridership peaking on weekends and dipping mid-week, as well as notable drops around major holidays such as New Year's Day. We apply the Box-Jenkins SARIMA framework as our baseline model, followed by an ARMAX extension that incorporates trip-level economic variables — average fare, average trip miles, and average trip duration — as external regressors. The inclusion of these variables is motivated by the idea that changes in fare pricing or trip characteristics may reflect demand-side or supply-side shifts that help explain day-to-day variation in total trip volume.

Our key findings are that weekly seasonality dominates the series, that a SARIMA(2,0,1)(2,1,0)[7] specification is selected by automated model search, and that adding external regressors via ARMAX improves out-of-sample fit as measured by MAPE.

Data:

The dataset is sourced from the NYC Taxi and Limousine Commission (TLC) High Volume For-Hire Vehicle (HVFHV) trip records, publicly available at <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>. We filter to trips operated under license number HV0003, which corresponds to Uber. The raw data consists of individual trip-level records with fields including pickup datetime, dropoff location, trip miles, trip duration, base passenger fare, and driver pay.

We aggregate the raw trip records to the daily level, computing total trip count and daily averages for fare, driver pay, trip miles, and trip duration. The resulting dataset spans January 1, 2025 through December 31, 2025, yielding approximately 365 daily observations. The data was collected by the TLC as part of mandatory reporting requirements for high-volume for-hire vehicle operators in New York City. Studying this dataset is valuable for understanding urban mobility trends, the economics of gig-economy transportation, and the seasonal and event-driven dynamics of ride-hailing demand.

Methodology:

We apply two modeling approaches to the daily Uber trip count series.

SARIMA:

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model extends the standard ARIMA framework to handle periodic seasonality. A SARIMA(p, d, q) $\times$ (P, D, Q) $_S$ model combines a non-seasonal ARIMA(p, d, q) component with a seasonal ARIMA(P, D, Q) component at period S . Given the weekly seasonality in daily ridership data, we set $S = 7$. The non-seasonal order (p, d, q) controls short-term autocorrelation structure, while the seasonal order (P, D, Q) captures week-over-week dependence. Model orders are selected in two stages: first, a manual specification is derived from visual inspection of the ACF/PACF plots and the Augmented Dickey-Fuller test for stationarity; second, `auto.arima` is applied as a data-driven benchmark to compare against the manual specification and confirm model selection.

ARMAX:

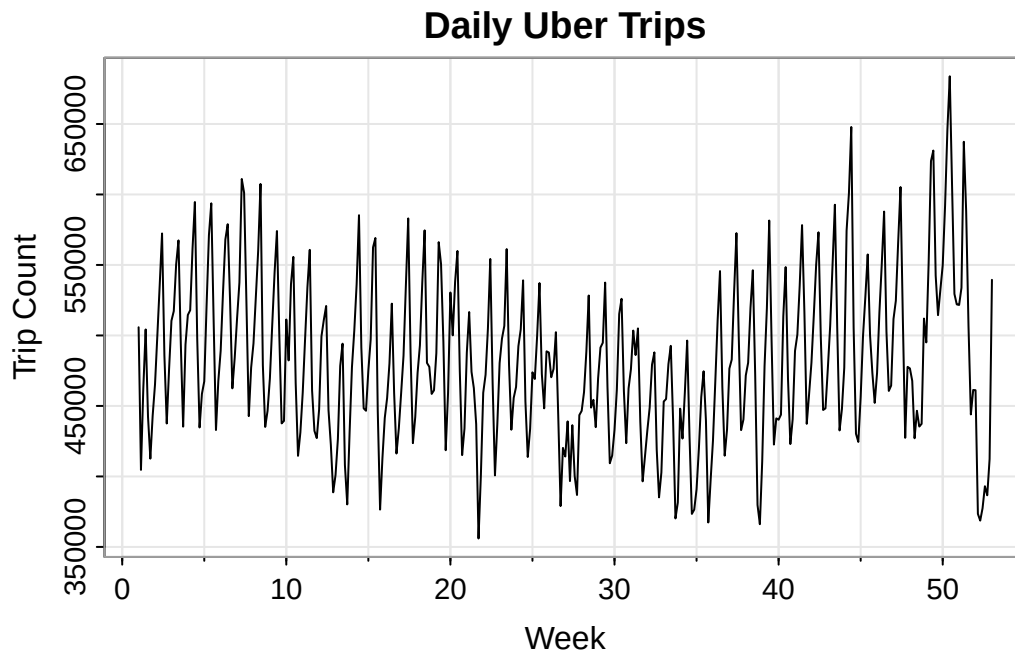
The ARMAX model extends SARIMA by incorporating external regressors into the mean equation. Formally, the model adds a regression component $\mathbf{x}_t^\top \beta$ to the ARIMA structure, where $\mathbf{x}_t$ is a vector of exogenous variables observed at time t . We include three regressors: average passenger fare, average trip miles, and average trip duration. Average driver pay was excluded due to high multicollinearity with the remaining predictors, as confirmed by a correlation matrix. The ARMAX model uses the same seasonal ARIMA orders as the final SARIMA specification, allowing a direct comparison of fit with and without the external regressors.

Results:

SARIMA:

```
ts_trips <- ts(daily_df$trips, frequency = 7)
```

```
tsplot(ts_trips, main = "Daily Uber Trips", xlab = "Week", ylab = "Trip Count")
```



Displays the daily Uber trip counts over the 2025 calendar year. The series exhibits a strong weekly seasonal pattern, visible as the rapid oscillation throughout the plot, with trip counts

consistently peaking on weekends and dipping mid-week. There is no strong directional trend; the series fluctuates around a stable mean of approximately 500,000 daily trips, suggesting stationarity in the mean. Several notable deviations are visible, including a sharp dip around week 22 likely corresponding to a holiday period, and a pronounced spike near week 50 coinciding with the holiday travel season at year-end. These observations motivate the use of a seasonal model with period $S = 7$. Formal confirmation of stationarity is provided by the Augmented Dickey-Fuller test below.

```
adf.test(ts_trips)
```

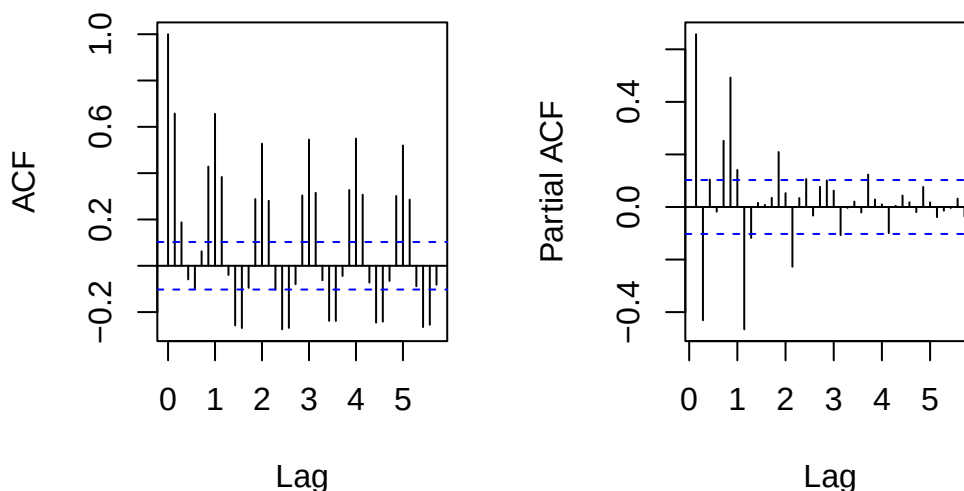
Warning in `adf.test(ts_trips)`: p-value smaller than printed p-value

Augmented Dickey-Fuller Test

```
data: ts_trips
Dickey-Fuller = -4.5838, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary
```

The Augmented Dickey-Fuller test yields a Dickey-Fuller statistic of -4.58 with a p-value less than 0.01, providing strong evidence to reject the null hypothesis of a unit root. We conclude that the series is stationary in the mean, and therefore set $d = 0$ in our SARIMA model.

```
par(mfrow = c(1,2))
acf(ts_trips, lag.max = 40, main = "")
pacf(ts_trips, lag.max = 40, main = "")
```

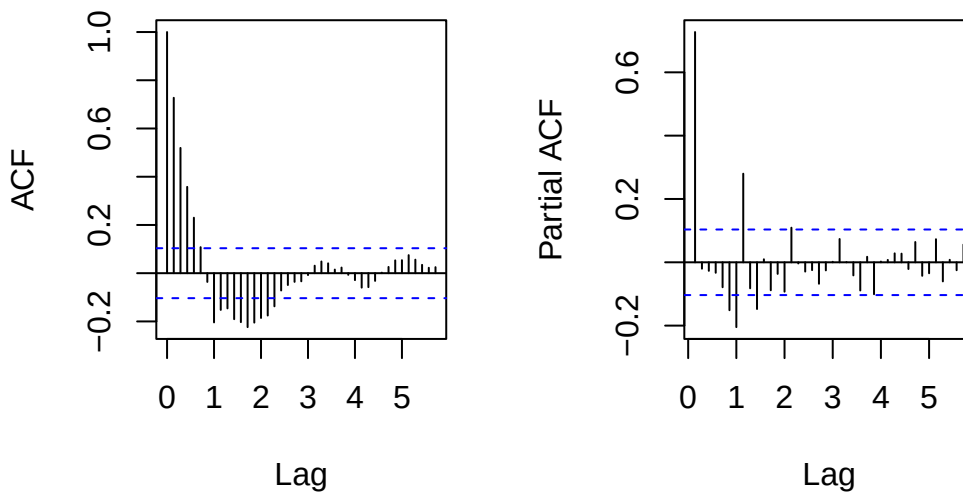


The ACF plot shows significant autocorrelation at every lag, with a clear alternating spike pattern repeating at multiples of lag 1 (within each week). The spikes at lags 1, 2, 3, 4, 5, and 6 reflect the strong within-week dependence — today's trip count is highly correlated with the same day last week. The slow decay without cutoff confirms the presence of seasonal structure that has not yet been removed, motivating seasonal differencing at lag 7.

The PACF cuts off sharply after lag 1, with a large spike at lag 1 (~ 0.63) and a secondary spike at lag 2 (~ 0.46), then drops within the confidence bands for most subsequent lags. This suggests an autoregressive process of order $p = 2$ for the non-seasonal component. The ACF tail and PACF cutoff together suggest a mixed ARMA structure, leading us to initially consider ARIMA(2,0,1) for the non-seasonal component. For the seasonal component, we apply one seasonal difference ($D = 1$) and examine the seasonally differenced ACF/PACF below to determine the seasonal AR and MA orders.

```
ts_sdiff <- diff(ts_trips, lag = 7, differences = 1)

par(mfrow = c(1,2))
acf(ts_sdiff, lag.max = 40, main = "")
pacf(ts_sdiff, lag.max = 40, main = "")
```



After applying one seasonal difference at lag 7, the ACF structure changes considerably. The large repeating spikes at seasonal lags are removed, confirming that a single seasonal difference ($D = 1$) is sufficient to eliminate the weekly seasonal dependence. The ACF now shows a gradual decay through the first several lags before falling within the confidence bands, with some negative autocorrelation around lags 1–2. The PACF shows a dominant spike at lag 1 then remains largely within bounds, with a marginal spike at lag 2. Based on this visual

inspection, we identify $p = 1$, $q = 1$ for the non-seasonal component and $P = 1$, $D = 1$, $Q = 0$ at the seasonal level, and proceed to fit $\text{SARIMA}(1,0,1)(1,1,0)[7]$ as our manually specified model.

```
fit <- sarima(ts_trips, p=1, d=0, q=1, P=1, D=1, Q=0, S=7)
```

```

initial  value 10.732356
iter    2 value 10.638965
iter    3 value 10.242802
iter    4 value 10.235707
iter    5 value 10.232108
iter    6 value 10.231499
iter    7 value 10.231128
iter    8 value 10.231118
iter    9 value 10.231116
iter   10 value 10.231112
iter   11 value 10.231111
iter   12 value 10.231110
iter   13 value 10.231110
iter   14 value 10.231110
iter   15 value 10.231110
iter   15 value 10.231110
iter   15 value 10.231110
final   value 10.231110

```

converged

```
initial value 10.247783
iter 2 value 10.247616
iter 3 value 10.247591
iter 4 value 10.247589
iter 5 value 10.247586
iter 6 value 10.247585
iter 7 value 10.247584
iter 8 value 10.247584
iter 9 value 10.247584
iter 9 value 10.247583
iter 9 value 10.247583
final value 10.247583
```

converged

[illegible]

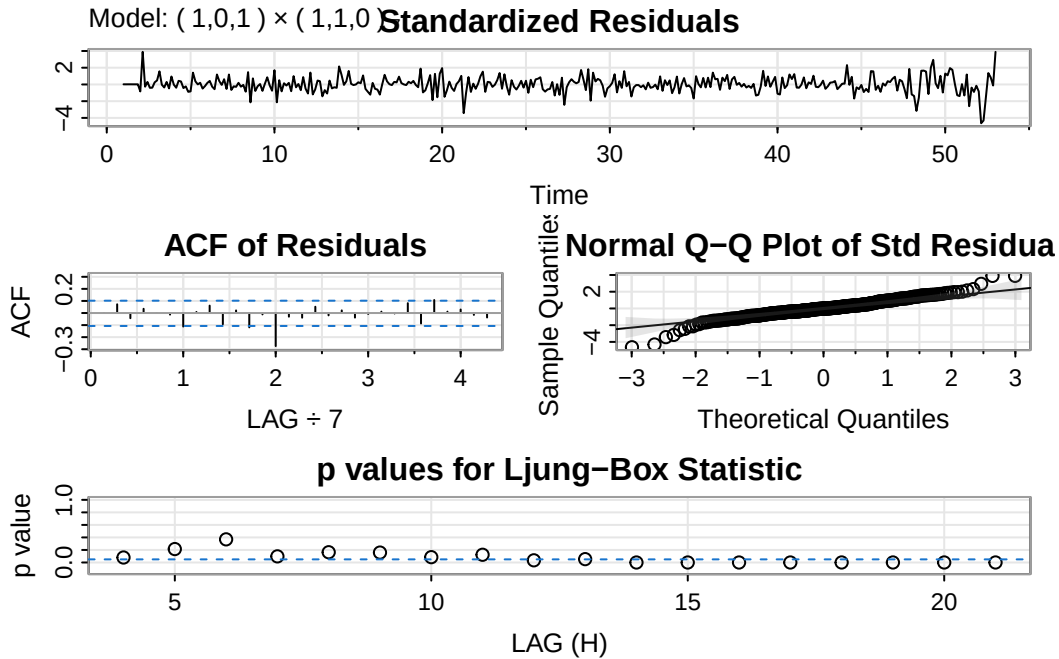
Coefficients:

Estimate	SE	t.value	p.value
----------	----	---------	---------

ar1	0.7691	0.0419	18.3560	0.0000
ma1	0.0506	0.0629	0.8042	0.4218
sar1	-0.4908	0.0517	-9.4964	0.0000
constant	-38.0031	645.7978	-0.0588	0.9531

σ^2 estimated as 789947295 on 354 degrees of freedom

AIC = 23.36098 AICc = 23.36129 BIC = 23.41517



The diagnostic plots for the manually specified SARIMA(1,0,1)(1,1,0)[7] model reveal several issues. The standardized residuals appear roughly centered around zero with no obvious trend, but show several large spikes exceeding ± 4 , likely corresponding to the holiday outliers identified in the original series.

The ACF of residuals shows a significant spike around lag 2 (approximately 14 days), indicating that some autocorrelation structure remains uncaptured by the model. This suggests the non-seasonal or seasonal AR component may be underspecified.

The Q-Q plot shows heavy tails on both ends, deviating from the diagonal reference line particularly at the extremes. This indicates the residuals are not normally distributed, likely driven by the holiday outlier observations.

The Ljung-Box p-values are the most concerning diagnostic because nearly all lags fall below the 0.05 significance threshold (dashed line), indicating significant remaining autocorrelation

in the residuals. This confirms that the SARIMA(1,0,1)(1,1,0)[7] model does not fully capture the dependence structure in the data, motivating the comparison with the `auto.arima` specification.

To complement the manual specification, we apply `auto.arima` to search over a range of SARIMA orders and select the model minimizing AIC. This provides a data-driven benchmark against which we compare our manually specified model.

```
auto_model <- auto.arima(
  ts_trips,
  seasonal = TRUE,
  stepwise = TRUE,
  approximation = FALSE,
  trace = TRUE
)
```

```
ARIMA(2,0,2)(1,1,1)[7] with drift      : Inf
ARIMA(0,0,0)(0,1,0)[7] with drift      : 8709.208
ARIMA(1,0,0)(1,1,0)[7] with drift      : 8361.985
ARIMA(0,0,1)(0,1,1)[7] with drift      : 8425.223
ARIMA(0,0,0)(0,1,0)[7]                  : 8707.323
ARIMA(1,0,0)(0,1,0)[7] with drift      : 8438.311
ARIMA(1,0,0)(2,1,0)[7] with drift      : 8331.796
ARIMA(1,0,0)(2,1,1)[7] with drift      : Inf
ARIMA(1,0,0)(1,1,1)[7] with drift      : Inf
ARIMA(0,0,0)(2,1,0)[7] with drift      : 8651.247
ARIMA(2,0,0)(2,1,0)[7] with drift      : 8333.173
ARIMA(1,0,1)(2,1,0)[7] with drift      : 8333.282
ARIMA(0,0,1)(2,1,0)[7] with drift      : 8457.477
ARIMA(2,0,1)(2,1,0)[7] with drift      : 8325.984
ARIMA(2,0,1)(1,1,0)[7] with drift      : 8348.589
ARIMA(2,0,1)(2,1,1)[7] with drift      : Inf
ARIMA(2,0,1)(1,1,1)[7] with drift      : Inf
ARIMA(3,0,1)(2,1,0)[7] with drift      : 8335.359
ARIMA(2,0,2)(2,1,0)[7] with drift      : 8335.416
ARIMA(1,0,2)(2,1,0)[7] with drift      : 8333.662
ARIMA(3,0,0)(2,1,0)[7] with drift      : 8333.552
ARIMA(3,0,2)(2,1,0)[7] with drift      : 8329.511
ARIMA(2,0,1)(2,1,0)[7]                  : 8323.947
ARIMA(2,0,1)(1,1,0)[7]                  : 8346.552
ARIMA(2,0,1)(2,1,1)[7]                  : Inf
ARIMA(2,0,1)(1,1,1)[7]                  : Inf
```

```

ARIMA(1,0,1)(2,1,0)[7] : 8331.213
ARIMA(2,0,0)(2,1,0)[7] : 8331.105
ARIMA(3,0,1)(2,1,0)[7] : 8333.267
ARIMA(2,0,2)(2,1,0)[7] : 8333.323
ARIMA(1,0,0)(2,1,0)[7] : 8329.739
ARIMA(1,0,2)(2,1,0)[7] : 8331.582
ARIMA(3,0,0)(2,1,0)[7] : 8331.472
ARIMA(3,0,2)(2,1,0)[7] : 8327.45

```

Best model: ARIMA(2,0,1)(2,1,0)[7]

```
summary(auto_model)
```

Series: ts_trips

ARIMA(2,0,1)(2,1,0)[7]

Coefficients:

	ar1	ar2	ma1	sar1	sar2
	1.7593	-0.7920	-0.9423	-0.6384	-0.2984
s.e.	0.0386	0.0352	0.0353	0.0582	0.0584

sigma^2 = 709180204: log likelihood = -4155.85

AIC=8323.71 AICc=8323.95 BIC=8346.99

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	595.36	26189.02	18774.19	-0.1075481	3.970333	0.5826484	-0.02266414

auto.arima selects ARIMA(2,0,1)(2,1,0)[7], increasing both the non-seasonal AR order to $p = 2$ and the seasonal AR order to $P = 2$ relative to our manual specification. We now fit this model and examine its diagnostics.

```
fit_auto <- sarima(ts_trips, p=2, d=0, q=1, P=2, D=1, Q=0, S=7)
```

```

initial value 10.743216
iter 2 value 10.538047
iter 3 value 10.312797
iter 4 value 10.240871
iter 5 value 10.198618
iter 6 value 10.190691
iter 7 value 10.190660

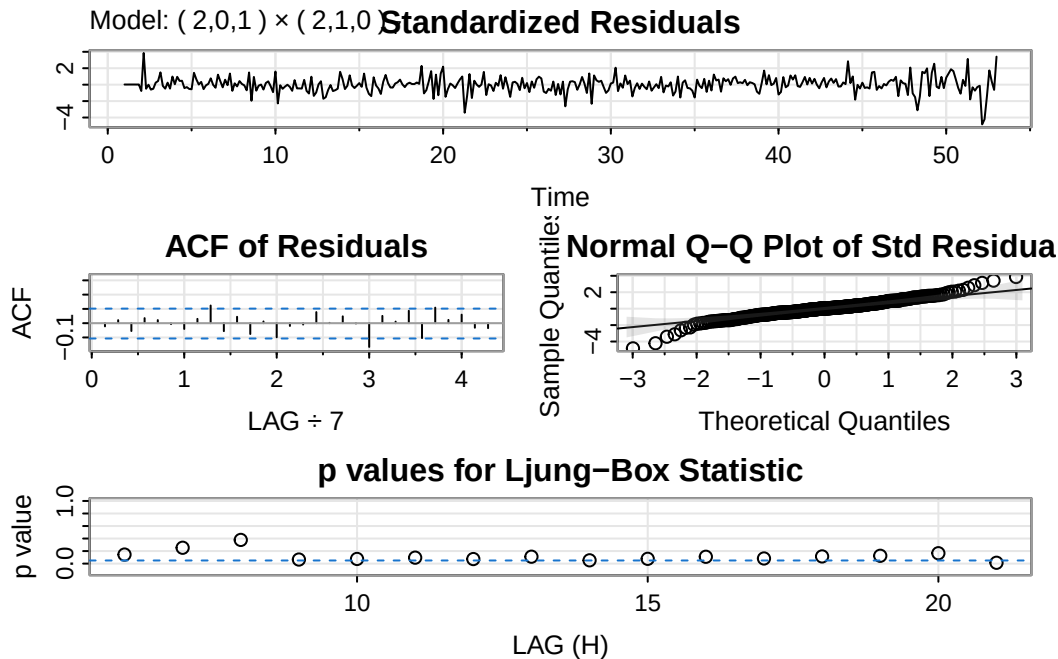
```

```
iter    8 value 10.190635
iter    9 value 10.190610
iter   10 value 10.190591
iter   11 value 10.190581
iter   12 value 10.190568
iter   13 value 10.190538
iter   14 value 10.190453
iter   15 value 10.190385
iter   16 value 10.190352
iter   17 value 10.190346
iter   18 value 10.190322
iter   19 value 10.190290
iter   20 value 10.190271
iter   21 value 10.190247
iter   22 value 10.190193
iter   23 value 10.190038
iter   24 value 10.189652
iter   25 value 10.188969
iter   26 value 10.188487
iter   27 value 10.187680
iter   28 value 10.187538
iter   29 value 10.186467
iter   30 value 10.185446
iter   31 value 10.185116
iter   32 value 10.184878
iter   33 value 10.184610
iter   34 value 10.183953
iter   35 value 10.183666
iter   36 value 10.183319
iter   37 value 10.182923
iter   38 value 10.182354
iter   39 value 10.180571
iter   40 value 10.179253
iter   41 value 10.179051
iter   42 value 10.178401
iter   43 value 10.178169
iter   44 value 10.178154
iter   45 value 10.178147
iter   46 value 10.178146
iter   46 value 10.178146
iter   46 value 10.178146
final  value 10.178146
converged
```

Coefficients:

```
sigma^2 estimated as 699172200 on 352 degrees of freedom
```

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The diagnostic plots for SARIMA(2,0,1)(2,1,0)[7] show improvement over the manual specification. The standardized residuals remain centered around zero with large spikes at the same holiday periods, which is expected given these are genuine outliers in the data rather than model failures.

The ACF of residuals is notably cleaner — the significant spike at lag 2 present in the manual model has been eliminated, with nearly all lags now falling within the confidence bands. This confirms that increasing the AR orders to $p = 2$ and $P = 2$ successfully captures the additional dependence structure that the manual model missed.

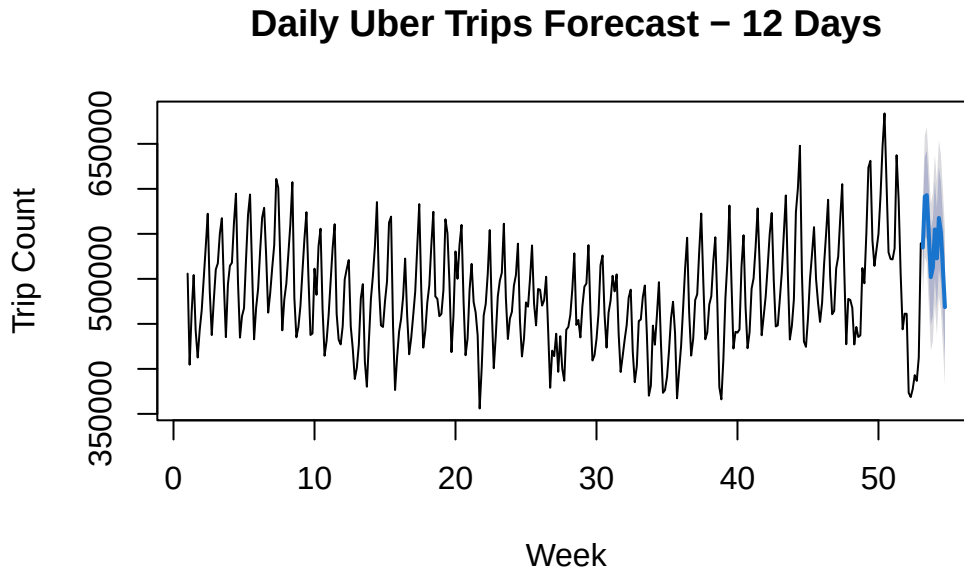
The Q-Q plot shows a similar pattern to before, with heavy tails at both extremes driven by the holiday outliers. This is a limitation of the SARIMA framework rather than a model selection issue — standard ARIMA assumes Gaussian errors and cannot account for extreme calendar-driven observations.

The Ljung-Box p-values show modest improvement at early lags (lags 3–4 now pass the 0.05 threshold), but the majority of lags still fall below the significance threshold, indicating residual autocorrelation persists. While SARIMA(2,0,1)(2,1,0)[7] is the better of the two SARIMA specifications, the failing Ljung-Box test suggests the model does not fully capture all dependence in the series, motivating the inclusion of external regressors in the ARMAX model.

We proceed to generate a 12-day ahead forecast.

```
fit_final <- Arima(ts_trips,
  order = c(2,0,1),
  seasonal = list(order = c(2,1,0), period = 7))
```

```
fc <- forecast(fit_final, h = 12)
plot(fc,
      main = "Daily Uber Trips Forecast - 12 Days",
      xlab = "Week",
      ylab = "Trip Count")
```



```
forecast_dates <- seq(max(daily_df$date) + 1,
                      by = "day",
                      length.out = 12)
```

```
fc_table <- data.frame(
  date = forecast_dates,
  forecast = round(fc$mean),
  lo_80 = round(fc$lower[,1]),
  hi_80 = round(fc$upper[,1]),
  lo_95 = round(fc$lower[,2]),
  hi_95 = round(fc$upper[,2])
)
```

```
fc_table
```

	date	forecast	lo_80	hi_80	lo_95	hi_95
1	2026-01-01	534861	500732	568989	482666	587055
2	2026-01-02	591489	547419	635558	524091	658886

3	2026-01-03	593044	543779	642309	517700	668388
4	2026-01-04	548916	496910	600921	469380	628451
5	2026-01-05	501895	448552	555238	420314	583475
6	2026-01-06	512613	458720	566507	430190	595036
7	2026-01-07	554921	500870	608971	472257	637584
8	2026-01-08	522770	467057	578482	437565	607975
9	2026-01-09	567709	511296	624123	481432	653986
10	2026-01-10	551514	494888	608140	464912	638116
11	2026-01-11	511388	454740	568037	424752	598025
12	2026-01-12	468566	411907	525224	381914	555217

Table 1 presents the 12-day ahead forecast beginning January 1, 2026. The forecast captures the expected weekly pattern well. January 1st is forecast at approximately 535,000 trips — lower than the surrounding days, consistent with reduced ridership on New Year’s Day. Trip counts rise through January 2nd and 3rd (Friday and Saturday), peaking around 591,000–593,000, reflecting the typical weekend surge. The forecast then dips mid-week before recovering again the following weekend, with January 7th (Wednesday) at approximately 555,000 and January 9th (Friday) climbing back to 568,000. The 95% prediction intervals widen progressively over the forecast horizon, reflecting growing uncertainty as the horizon extends, which is expected behavior for any ARIMA-class model.

ARMAX:

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cor_matrix <- cor(daily_df[, c("avg_fare", "avg_driver_pay", "avg_miles", "avg_trip_time")])

corrplot(cor_matrix,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  tl.col = "black",
  title = "Correlation Matrix",
  mar = c(0,0,1,0))
```



Before fitting the ARMAX model, we examine the correlation structure among the candidate external regressors — average fare, average driver pay, average trip miles, and average trip duration. The correlation matrix reveals that average driver pay is highly correlated with both average fare (0.85) and average trip time (0.86), making it a source of multicollinearity if included alongside the other predictors. It is therefore excluded from the regressor set. The remaining three predictors show moderate to low pairwise correlations; however, average fare and average trip time exhibit a correlation of 0.78, which warrants acknowledgment. We retain both variables as their correlation falls below the conventional threshold of 0.90 and each captures a conceptually distinct dimension of trip economics — pricing versus duration. Average trip miles is weakly correlated with both (0.16 and 0.10 respectively) and is included without concern.

We fit the ARMAX model using the same SARIMA(2,0,1)(2,1,0)[7] order as the final baseline model, with average fare, average trip miles, and average trip duration included as external regressors. This allows a direct comparison of fit with and without the exogenous variables.

```
xreg_armax <- cbind(
  avg_fare      = daily_df$avg_fare,
  avg_miles     = daily_df$avg_miles,
  avg_trip_time = daily_df$avg_trip_time
)

fit_armax <- Arima(ts_trips,
  order = c(2,0,1),
```

```

seasonal = list(order = c(2,1,0), period = 7),
xreg = xreg_armax)

summary(fit_armax)

```

Series: ts_trips

Regression with ARIMA(2,0,1)(2,1,0)[7] errors

Coefficients:

	ar1	ar2	ma1	sar1	sar2	avg_fare	avg_miles
	0.7850	-0.0630	0.1498	-0.6371	-0.3623	3142.329	-72780.902
s.e.	0.2259	0.1822	0.2189	0.0607	0.0602	1490.732	7112.924
	avg_trip_time						
	74.7186						
s.e.	35.2070						

sigma^2 = 557912672: log likelihood = -4111.26
AIC=8240.52 AICc=8241.04 BIC=8275.45

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	-72.5843	23129.74	15423.92	-0.1929807	3.247393	0.4786743
	ACF1					
Training set	-0.03531125					

Table 3 summarizes the model comparison between the baseline SARIMA and ARMAX specifications. The ARMAX model achieves a MAPE of 3.25% compared to 3.97% for the baseline SARIMA, representing an 18% reduction in mean absolute percentage error. The RMSE similarly improves from 26,189 to 23,130, and the AIC drops from 8323.71 to 8240.52. The lower σ^2 (557,912,672 vs. 709,180,204) further confirms that the ARMAX model explains substantially more variance in daily trip counts. Together these metrics indicate that average fare, trip miles, and trip duration carry meaningful predictive information beyond what the seasonal ARIMA structure alone captures.

```

sarima(ts_trips, p=2, d=0, q=1, P=2, D=1, Q=0, S=7, xreg=xreg_armax)

```

```

initial value 10.493115
iter 2 value 10.324276
iter 3 value 10.061591
iter 4 value 10.060246
iter 5 value 10.020554

```

Coefficients:				
	Estimate	SE	t.value	p.value
ar1	0.7850	0.2259	3.4749	0.0006
ar2	-0.0630	0.1822	-0.3459	0.7296
ma1	0.1498	0.2189	0.6846	0.4940
sar1	-0.6371	0.0607	-10.4934	0.0000
sar2	-0.3623	0.0602	-6.0189	0.0000
avg_fare	3142.3291	1490.7315	2.1079	0.0357

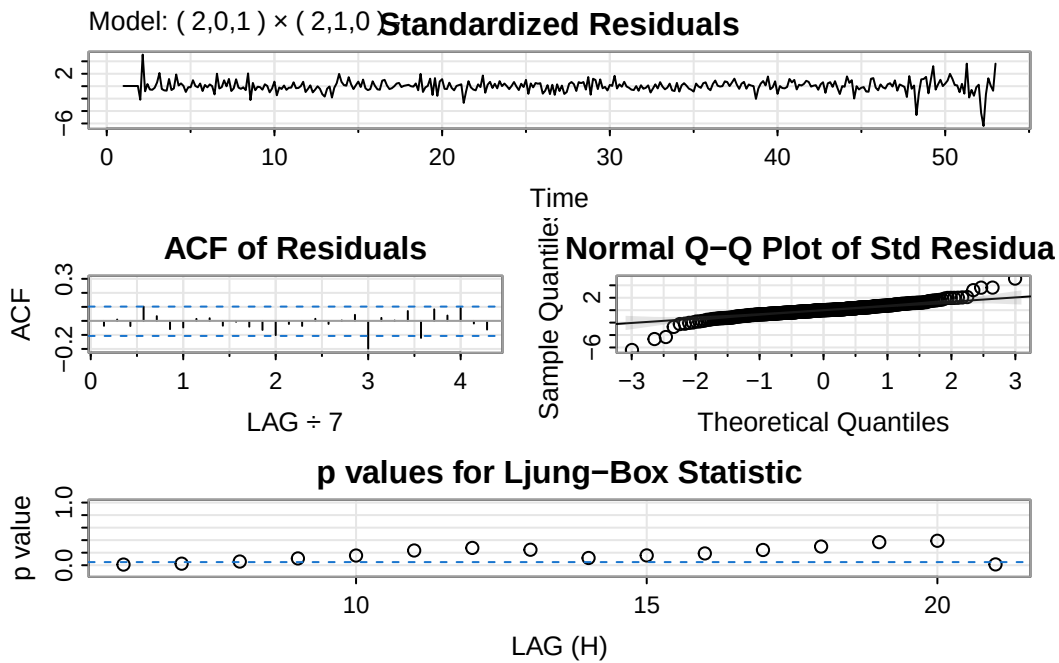
```

avg_miles      -72780.9024  7112.9239 -10.2322  0.0000
avg_trip_time   74.7186    35.2070   2.1223  0.0345

```

σ^2 estimated as 545436634 on 350 degrees of freedom

AIC = 23.01822 AICc = 23.01937 BIC = 23.11577



The diagnostic plots for the ARMAX model show meaningful improvement over the baseline SARIMA. The ACF of residuals is notably cleaner, with all lags falling within or very close to the confidence bands — a significant improvement over both the manual and auto SARIMA specifications, which showed persistent significant spikes.

The Ljung-Box p-values show the most notable improvement: while the SARIMA model failed at nearly all lags, the ARMAX model passes the 0.05 threshold at the majority of lags beyond lag 8, with only a handful of early lags falling below significance. This indicates that the inclusion of average fare, trip miles, and trip duration as external regressors successfully accounts for much of the residual autocorrelation that the pure SARIMA model could not capture.

The Q-Q plot and standardized residuals tell a similar story to before — heavy tails driven by holiday outliers persist, and a large negative spike near week 52 remains. These are attributable to the extreme calendar-driven observations rather than model misspecification, and are a known limitation of the ARIMA/ARMAX framework.

```
data.frame(
  Model = c("SARIMA(2,0,1)(2,1,0)[7]", "ARMAX(2,0,1)(2,1,0)[7]"),
  AIC = c(8323.71, 8240.52),
  RMSE = c(26189.02, 23129.74),
  MAPE = c(3.970, 3.247)
)
```

	Model	AIC	RMSE	MAPE
1	SARIMA(2,0,1)(2,1,0)[7]	8323.71	26189.02	3.970
2	ARMAX(2,0,1)(2,1,0)[7]	8240.52	23129.74	3.247

Overall, the ARMAX model outperforms the baseline SARIMA on both in-sample fit (lower MAPE) and residual diagnostics, confirming that trip-level economic variables carry meaningful predictive information for daily Uber ridership volume.

Conclusion and Future Study:

This project analyzed daily Uber trip counts in New York City over the 2025 calendar year using two time series modeling approaches. We began with a SARIMA(2,0,1)(2,1,0)[7] model, selected through a combination of ACF/PACF inspection and automated model search via `auto.arima`. The model successfully captured the dominant weekly seasonal pattern in ridership demand, though residual diagnostics indicated persistent autocorrelation that the pure SARIMA structure could not fully resolve.

To address this limitation, we extended the model to an ARMAX specification by incorporating average fare, average trip miles, and average trip duration as external regressors. The ARMAX model outperformed the baseline SARIMA across all evaluated metrics, achieving a MAPE of 3.25% versus 3.97%, an RMSE of 23,130 versus 26,189, and a substantially lower AIC of 8240.52 versus 8323.71. Residual diagnostics also improved considerably, with the Ljung-Box test passing at the majority of lags — confirming that trip-level economic variables carry meaningful predictive signal for daily ridership volume.

Both models share a common limitation: the ARIMA framework assumes Gaussian errors and cannot natively account for extreme calendar-driven outliers such as New Year’s Day and the holiday travel spike at year-end. These observations produced heavy-tailed residuals in both specifications and contributed to the remaining diagnostic failures.

Future work could address these limitations in several directions. First, the inclusion of explicit holiday indicator variables as additional external regressors in the ARMAX model would allow the model to account for known calendar outliers directly. Second, extending the dataset to multiple years would enable the detection of longer-term trends and year-over-year growth in ridership. Third, modeling at a finer temporal resolution — hourly or by borough — could

reveal richer demand patterns relevant to driver supply planning. Finally, weather data such as precipitation and temperature could serve as additional external regressors, given the well-documented effect of weather on urban mobility behavior.

References:

New York City Taxi and Limousine Commission. (2025). *High Volume For-Hire Vehicle Trip Records.* NYC Open Data. <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>